Supplementary Material
for
Can triage with qualitative models make MSE more efficient?

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Appendix A: The Operating Model

A.1 Basic population dynamics

The dynamics of each modelled species are governed by a multi-area age-structured model, i.e.:

$$N_{y,a}^{A} = \begin{cases} \chi^{A} R_y & \text{if } a = 0 \\ \sum_{A'} \Omega_{y-1,a-1}^{A',A} N_{y-1,a-1}^{A} \, e^{-Z_{y-1,a-1}^{A'}} & \text{if } 1 < a < x \\ \sum_{A'} \Omega_{y-1,x-1}^{A',A} N_{y-1,x-1}^{A} \, e^{-Z_{y-1,x-1}^{A'}} + \sum_{A'} \Omega_{y-1,}^{A',A} N_{y-1,x}^{A} \, e^{-Z_{y-1,x}^{A'}} & \text{if } a = x \end{cases} \quad \text{(A.1)}$$

where $N_{y,a}^{A}$ is the number of animals of age *a* in region *A* ("inshore", "offshore" and "far offshore") at the start of year *y*, R_y is the total recruitment (at age-0) to the population during year *y*, χ^{A} is the proportion of recruitment to region *A* (0 except for the "inshore" area, which is 1), $Z_{y,a}^{A}$ is the total mortality on animals of age *a* in region *A* during year *y*:

$$Z_{y,a}^{A} = M + \sum_{f \in A} S_a^f F_y^f \quad \text{(A.2)}$$

M is the rate of natural mortality, S_a^f is the selectivity of fishery *f* on animals of age *a*, F_y^f is the fully-selected fishing mortality by fishery *f* during year *y*, $f \in A$ denotes that fishery *f* takes place in region *A*, and $\Omega_{y,a}^{A',A}$ is the proportion, during year *y*, of animals of age *a* that move from region *A'* to region *A* at the end of the year (after natural and fishing mortality), and *x* is the maximum age (assumed to be a plus-group).

A.2 Recruitment and spawning

Recruitment is related to spawning biomass according to a Ricker stock-recruitment relationship, i.e.:

$$R_y = \tilde{\alpha}_y \alpha \, R_0 \, \tilde{S}_y / \tilde{S}_0 \, e^{\ln(5h)\left(1-\frac{\tilde{S}_y}{\tilde{S}_0}\right)/0.8} e^{\varepsilon_y - \sigma_R^2/2} \qquad \varepsilon_y \sim N(0, \sigma_R^2) \quad \text{(A.3)}$$

where R_0 is unfished recruitment, $\tilde{S}_y$ is the spawning biomass at the start of year *y*:

$$\tilde{S}_y = \sum_A \sum_{a>1} w_{y,a} m_a N_{y,a}^{A} \quad \text{(A.4)}$$

$w_{y,a}$ is the weight of animal of age *a* at the start of year *y*, m_a is the proportion of animals of age *a* that are mature, $\tilde{S}_0$ is unfished spawning biomass, *h* is the "steepness" of the stock-recruitment relationship (the proportion of unfished recruitment expected when spawning biomass is reduced to 20% of the unfished level; Francis, 1992), $\tilde{\alpha}_y$ is a factor to account for climate-related factors on expected recruitment ($\tilde{\alpha}_y = 1$ unless there are climate effects on recruitment), and σ_R is the standard deviation of the random deviations in recruitment about the stock-recruitment relationship.

We also ran our analyses relating recruitment to spawning biomass according to a Beverton-Holt stock-recruitment relationship, i.e.:

$$R_y = \frac{\tilde{\alpha}_y 4h \, R_0 \, \tilde{S}_y / \tilde{S}_0}{(1-h)+(5h-1)\tilde{S}_y/\tilde{S}_0} \, e^{\varepsilon_y - \sigma_R^2/2} \qquad \varepsilon_y \sim N(0, \sigma_R^2) \quad \text{(A.5)}$$

(A.6)

Length-at-age is modelled by:

$$L_{y,a} = \begin{cases} L_\infty\left(1 - e^{\kappa_1(a-t_0)}\right) & \text{if } y = 1 \\ L_{y-1,a-1} + \left(L_\infty - L_{y-1,a-1}\right)\left(1 - e^{\kappa_y(a-t_0)}\right) & \text{if } y > 1 \end{cases} \quad \text{(A.7)}$$

where L_∞ and t_0 are two of the parameters of the von Bertlanffy growth curve, and κ_y is the value of the von Bertalanffy growth rate parameter during year *y*. Weight-at-age is given by $\delta_1(L_{y,a})^{\delta_2}$ where δ_1 and δ_2 are the length-weight parameters.

The parameters for the two species are given in Table A.1.

A.3. Catches

The catch in weight during year *y* from area *A*, $\tilde{C}_y^A$, is given by:

$$\tilde{C}_y^A = \sum_{f \in A} \sum_a w_{y,a+0.5}\, C_{y,a}^f \quad \text{(A.8)}$$

where $C_{y,a}^f$ is the catch of animals of age *a* by fishery *f* during year *y*:

$$C_{y,a}^f = \frac{S_a^f F_y^f}{Z_{y,a}^A} N_{y,a}^A \left(1 - e^{-Z_{y,a}^A}\right) \quad \text{(A.9)}$$

and area *A* is the region in which fishery *f* operates.

A.3 Fishing mortality

The status-quo relative fishing mortality rates between fisheries is 1/6, 2/6, and 1/2. The fishing patterns by fishery are subject to annual log-normal error (CV=0.3) so that the time-trajectory of fishing mortality differs somewhat among replicate simulations.

The baseline assumption regarding future fishing mortality is that effort (and hence fishing mortality) is constant over time. For management strategy C, the effort for fishery 1 increases by 50% when it is no longer allowed to fish in the "inshore" region.

Management strategy 1 involves computing an overall target fishing mortality according to Fig. 1f and rescaling the fishing mortality rates so that the catch according to Equation A.7 matches the expected catch given the target fishing mortality. Operating model 1 involves calculating F_y^f according to:

$$F_y^f = F_{y-51}^f \left(B_y^{E,f} / \left(\tau_{51} B_{51}^{E,f}\right)\right)^{\tau_2} \quad \text{(A.10)}$$

where $B_y^{E,f}$ is the exploitable biomass during year *y* for fishery *f*:

$$B_y^{E,f} = \sum_a \sum_{A \in f} w_{y,a}\, S_a^f N_{y,a}^A \left(1 - e^{-Z_{y,a}^A}\right) / Z_{y,a}^A \quad \text{(A.11)}$$

$\tau_1 = 1$, $\tau_1 = 0.9$, which leads to the exploitable biomass equilibrating at the exploitable biomass in the first year of the projection period with damped oscillation.

A.4 Effects of climate on the population dynamics

The base-case analysis assumes that there are no climate effects on movement, growth and recruitment. The climate-related scenarios are:

- Expected recruitment changes over time. This is modelled by halving the value of $\widetilde{\alpha}_y$ from year 51 to year 60 (Fig. 3c).
- Time-varying growth is modelled by allowing κ_y to change linearly between year 51 and the end of the projection period) (Fig. 3a).
- Movement changes over time to allow some animals to move to the "far offshore" area (area 3) such that in year 51, there is a probability of M_p of animals (independent of age) moving from the 'offshore" to "far offshore" region. The probability of animals moving from the "far offshore" to the "offshore" region is 0.05.

A.5 Data generation

The monitoring data available for assessment purposes (only used for management strategy 1) are the catches by area, catch age-composition data by fishery, and survey index and age-composition data by area. The survey index is generated according to:

$$I_y^A = \sum_a w_{y,a}\, S_a^{s,A}\, N_{y,a}^A e^{\eta_y - \sigma_I^2/2} \qquad \eta_y \sim N(0, \sigma_I^2) \qquad \text{(A.12)}$$

where $S_a^{s,A}$ is survey selectivity for region *A*, and σ_I is the standard error of the logarithm of the survey index of biomass. The survey age-composition data are Dirichlet samples based on expected values proportional to $S_a^{s,A} N_{y,a}^A$. The catches are assumed to be known exactly while the catch proportion data for fishery *f* are Dirichlet samples with expected values proportional to the catch-at-age data for fishery *f* (Eqn A.7).

The fishery age-composition data by fishery are assumed to be available for the years that the fishery operates (year 25 for fishery 1 and year 30 for fishery 2), with an effective sample size of 100. The survey indices are assumed to be available annually from year 31. Assessments and hence updates to the catch limit occur every five years.

The catch limit from the assessment method is computed using the harvest control rule (Fig. 1f). The fully-selected fishing mortality rates for projecting the operating model forward are then computed by solving for fishing mortality rates given the total catch over all fleets equals the catch limit.

A.6. References

Francis, R.I.C.C., 1992. Use of risk analysis to assess fishery management strategies: A case study using orange roughy (*Hoplostethus atlanticus*) on the Chatham Rise, New Zealand. Can. J. Fish. Aquat. Sci. 49, 922-930.

A.7 Tables and Figures

Table A.1.Values for the parameters of the age-structured operating model.

Parameter	Species 1	Species 2
Maximum age, x	20	8
Growth		
Length-at-age	Fig. 1a	Fig. 1a
Weight-length relationship, δ_1	1	1
Weight-length relationship, δ_2	3	3
Stock-recruitment relationship		
Fecundity	Fig. 1b	Fig. 1b
Steepness	0.7	0.8
σ_R	0.6	1.0

R_0	100	100
Base-case movement (from inshore to offshore)	0 for ages 0 and 1; linear increase from 0 at age 1 to 0.2 at age 5.	0 for ages 0 and 1; linear increase from 0 at age 1 to 0.1 at age 4.
Fishery selectivity	Fig. 1c	Fig. 1d

Table A2. Results from the qMSE operating model 1 (Ricker stock-recruit relationship), comparing the main text results using a target exploitable biomass of 1.0x biomass, with the alternative values of 0.75x and 1.25x biomass. qMSE results are reported as the probability that each output metric increases relative to management strategy A (strategies B-E). The color of the cell indicates whether there was a > 60% (green) versus < 40% (orange) probability of increase. For comparison to the QNMs, equivocal outcomes (40% < x < 60%) are in uncolored cells.

To the right of the qMSE predictions for each species, we sum the qMSE predictions for the given case x target biomass that matched ("Total match") or mismatched ("Total mismatch") the corresponding QNM predictions. For some variables, QNM predictions were equivocal, and therefore not compared ("QNM equivocal"). There was one case (1-C) for which the total match / mismatch to QNMs differed between alternate implementations of the qMSE operating model 1; however, the totals differed only by a count of 1.

		Species 1				Species 2						
		Catch				Catch						
Case	Target Biomass	Total	Fishery 1	Fishery 2	Spawning biomass	Total	Fishery 1	Fishery 2	Spawning biomass	Total match	Total mismatch	QNM equivocal
1-B	0.75x	100	0	100	99.5	38	0	99.5	97.5	6	0	2
	1.0x	100	100	0	37	47	56	0	90.5	6	0	2
	1.25x	100	100	100	100	43	35.5	99	97.5	6	0	2
1-C	0.75x	100	99.5	100	100	38	27	98.5	97.5	4	4	0
	1.0x	96	0	100	93	35.5	*0*	99.5	89	5	3	0
	1.25x	94	100	0	5	58	*68.5*	0	70	4	4	0
1-D	0.75x	98.5	84	100	99.5	36	*28*	97	92.5	0	0	8
	1.0x	98.5	81.5	100	99.5	32.5	23.5	96	91	0	0	8
	1.25x	65.5	0	100	56.5	34	0	99.5	75.5	0	0	8
1-E	0.75x	90	100	0	0.5	71.5	82.5	0	45	5	1	2
	1.0x	95.5	80.5	100	93.5	33.5	28	88.5	80.5	5	1	2
	1.25x	93.5	79	99.5	85.5	30.5	25	89.5	79	5	1	2

Table A3. Matches and mismatches in outcomes between the quantitative MSE and the corresponding QNMs for operating model scenario 1 (Thorson effort; Ricker stock-recruit relationship). We present the proportion of comparisons (4 metrics x 4 management strategies) for which the direction of change (positive / negative) for a given metric was the same as in the qMSE. We then report the proportion which did not match, but the QNM output was equivocal (i.e., <60% sign consistency; "Mismatch (equivocal QNM)"), and the proportion which did not match with non-equivocal QNM output ("Mismatch"). We also list the management strategy code x metric combinations for which there were directional mismatches ("Mismatched HCRs"). The main text used a qMSE in which the target was 1x biomass.

	Species 1: medium-longevity			Species 2: short-lived		
	Target 0.75x Biomass	Target 1x Biomass (main text)	Target 1.25x Biomass	Target 0.75x Biomass	Target 1x Biomass (main text)	Target 1.25x Biomass
Match:						
Only non-equivocal QNMs	0.50	0.50[+]	0.50[+]	0.44[+]	0.50	0.44
Only equivocal QNMs	0.31	0.31	0.31	0.19	0.19	0.19
All QNMs	*0.81*	*0.81*	*0.81*	*0.63*	*0.69*	*0.63*
Mismatch:						
Only non-equivocal QNMs	0.13	0.13	0.13	0.19	0.13	0.19[+]
Only equivocal QNMs	0.06	0.06	0.06	0.19[+]	0.19	0.19
All QNMs	*0.19*	*0.19*	*0.19*	*0.38*	*0.32*	*0.38*

[+]Includes 1 case in which the qMSE outcome was equivocal.

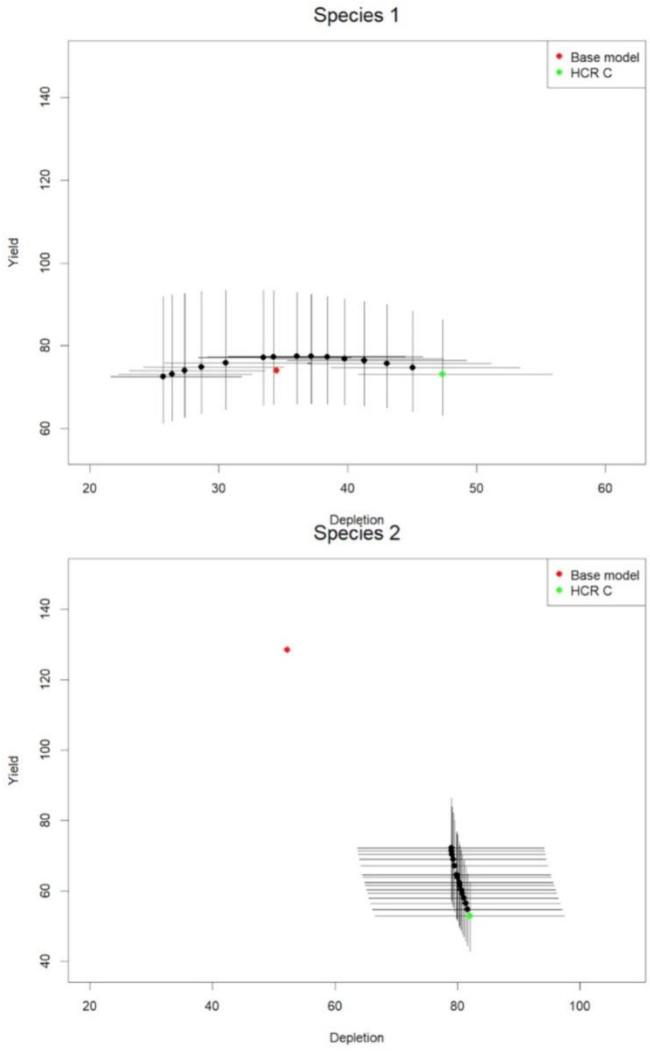


Figure A.1. Total catch (y axis, 'yield') plotted against stock depletion (x axis, 'Depletion') when the baseline operating model (OM 0) is run with different values of offshore fishing effort. The red point in each graph represents the output from the base model, OM 0 under HCR A (Status Quo), and the green point represents output from OM 0 under HCR C (Fishery 1 effort in the inshore region is set to zero, and the effort for Fishery 1 in the offshore region increases by 50%). Intermediate black points represent output from OM 0 while varying the increase in Fishery 1 offshore effort between 0% and 50% (i.e., management strategy or "HCR" C). This shows that the depletion of Species 2 is relatively consistent despite increases in offshore fishing effort, whereas Species 1 becomes markedly more depleted as Fishery 1 offshore effort increases. This is because the majority of Species 2 biomass resides in the inshore area, and is unaffected by changing fishing effort in the offshore area.

Appendix B: The QNMs

B.1. From the main text: Description of QNMs

Changes to each QNM from the base operating model scenario are described in detail in Tables B1-3; Figs B1-3, but in short:

1. Operating model scenario 1 (QNM 1): There are additional links between biomass and fishing effort.
2. Operating model scenario 2: There is a negative perturbation to *recruits.*
3. Operating model scenario 3 (QNM 3): There are new variables to represent a far offshore location, and a positively perturbed *climate pressure* that drives offshore movement
4. Operating model scenario 4 (QNM 4): There is a new variable *growth*, which is linked to biomass and is positively perturbed.

Additional structural modifications and/or perturbations of fishing effort and mesh size variables were needed to explore management strategies B - E (Table B3).

B.2. Tables and Figures

Table B.1. Summary of QNM complexity for each operating model scenario x management strategy combination. Variable count (a) is the total number of nodes or vertices in the network; link density (b) is the proportion of total possible links that are present in the network; and the number of perturbations (c) accounts for perturbations used to represent both the management strategy and the operating model scenario (see Tables B2, B3).

(a) Variable Count

Operating Model Scenario	**All Management Strategies**
0. Status Quo	20
1.Effort scales according to the Thorson effort model	20
2.Climate reduces productivity	20
3.Movement into the “far offshore” region	23
4.Growth rate increases in response to climate change	21

(b) Link Density

	Management Strategy				
Operating Model Scenario	**A**	**B**	**C**	**D**	**E**

(b) Link Density

0. Status Quo	0.168	0.150	0.161	0.168	0.158
1.Effort scales according to the Thorson effort model	0.158	0.147	0.155	0.158	0.153
2.Climate reduces productivity	0.168	0.150	0.161	0.168	0.158
3.Movement into the "far offshore" region	0.154	0.140	0.148	0.154	0.146
4.Growth rate increases in response to climate change	0.160	0.143	0.152	0.160	0.150

(c) Number of Perturbed Variables

	Management Strategy					
Operating Model Scenario	**A**	**B**	**C**	**D**	**E**	***Mean*** *(± 0.84)*
0. Status Quo	0	2	1	2	1	*1.2*
1.Effort scales according to the Thorson effort model	0	2	1	2	1	*1.2*
2.Climate reduces productivity	1	3	2	3	2	*2.2*
3.Movement into the "far offshore" region	4	6	5	6	5	*5.2*
4.Growth rate increases in response to climate change	2	4	3	4	3	*3.2*
Mean *(± 1.7)*	*1.4*	*3.4*	*2.4*	*3.4*	*2.4*	

Table B.2. Changes to the baseline operating model scenario (QNM 0; Fig 5a main text) under the status quo management strategy, for each of the other operating model scenarios.

MSE Operating Model Scenario	**QNM ID**	**Structural change from baseline scenario (QNM 0)**	**Perturbation**	**Intent**
1.Effort scales according to the Thorson effort model	1	Add Links. (1,2) Inshore biomass small / large --> inshore small mesh effort		Completes a feedback loop between fishing activity and biomass. As the targeted biomass changes, fishing effort will

		(3,4) Offshore biomass small / large --> offshore small mesh effort (5) Offshore biomass large --> offshore large mesh effort Remove Links. Inshore/offshore biomass small/large --> Total / Fishery 1 / Fishery 2 catch *(See Fig. 5b in main text)*		change; this will influence fishing activity variables, catch, and biomass itself. Because of this, we don't need the additional links from biomass to total catch.
2.Climate reduces productivity	2	None.	(-)Recruits	Negative impact on recruits will reduce biomass
3.Movement into the "far offshore" region	3	Add Variables. (1) Climate pressure (2) Far offshore biomass small (3) Far offshore biomass large Add Links. (1) Inshore biomass small --> far offshore biomass small (2) Inshore biomass large --> far offshore biomass large (3,4) Climate pressure --* Offshore biomass small/large (5,6) Climate pressure --> Far offshore biomass small/large *(See Fig. 5c in main text)*	(+) Climate pressure	Create relationship between inshore biomass and far offshore biomass, to represent movement. As there are more fish in an far offshore region, the offshore biomass available to fisheries will be reduced. Perturbation "activates" impacts we expect from movement into an inaccessible location
4.Growth rate increases in response to climate change	4	Add Variable. (1)Growth Add Links. (1,2) Growth --> Inshore/Offshore biomass small *(See Fig. 5d in main text)*	(+)Growth	Faster growth will make smaller fish bigger, creating more biomass. Perturbation "activates" impacts to growth.

Table B.3. Changes to any given QNM under the status quo management strategy (Table B2) to represent management strategies B – E.

	Operating Model Scenario	**0**	**1**	**2**	**3**	**4**
Management strategy						
B. Eliminate fishery 1, which	Structural change (Fig. B1)	Remove Links (n=3). Small fish / inshore large fish biomass --> total / fishery 1 catch	Remove Links (1-4 in Table B1). biomass --> inshore small mesh effort	- -	- -	- -

catches small fish	Perturbation	(-)inshore small mesh effort (-)offshore small mesh effort	--	- -	- -	- -
C. Constrain fishery 1 to the "offshore" area, increasing the effort (and fishing mortality) level there	Structural change (Fig. B2)	Add Link (n=1). Inshore small mesh effort --* offshore small mesh effort Remove Links (n=2). Inshore fish biomass --> total / fishery 1 catch	Add Link (n=1). Inshore small mesh effort --* offshore small mesh effort Remove Links. (1-2 in Table B1) inshore biomass --> small mesh effort	- -	- -	- -
	Perturbation	(-)inshore small mesh effort	--	- -	- -	- -
D. Increase the mesh size of fishery 1	Structural change	--	--	- -	- -	- -
	Perturbation	(+)inshore mesh size (+)offshore mesh size	--	- -	- -	- -
E. Eliminate fishery 1 from the "inshore" area, without a change in effort in the offshore areas	Structural change (Fig. B3)	Remove Links (n=2). Inshore fish fish biomass --> total / fishery 1 catch	Remove Links. (1-2 in Table B1) inshore biomass --> small mesh effort	- -	- -	- -
	Perturbation	(-)inshore small mesh effort	--	- -	- -	- -

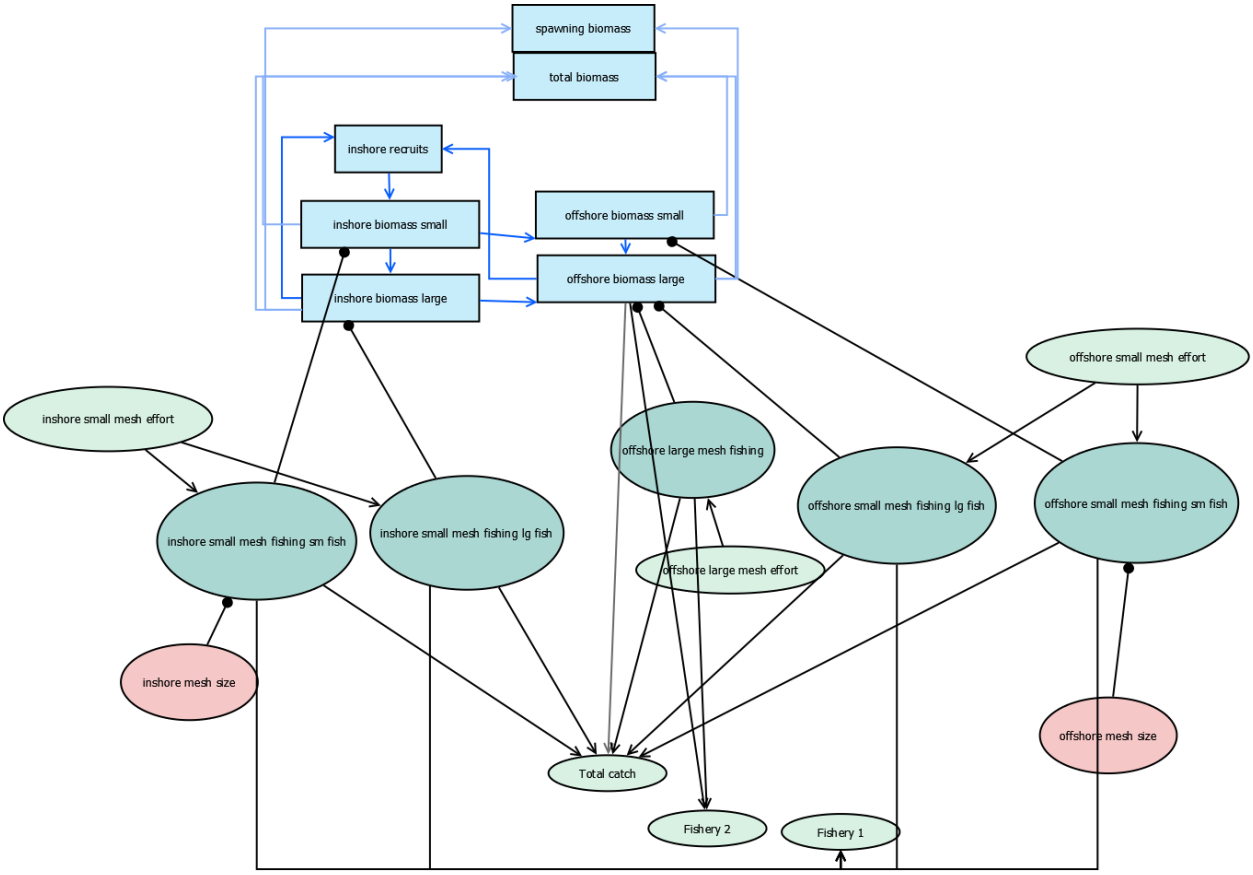


Fig B.1. Management strategy B implemented in the baseline operating model scenario (QNM 0). Incoming links to each performance metric are the same as in Figure 4 in the main text.

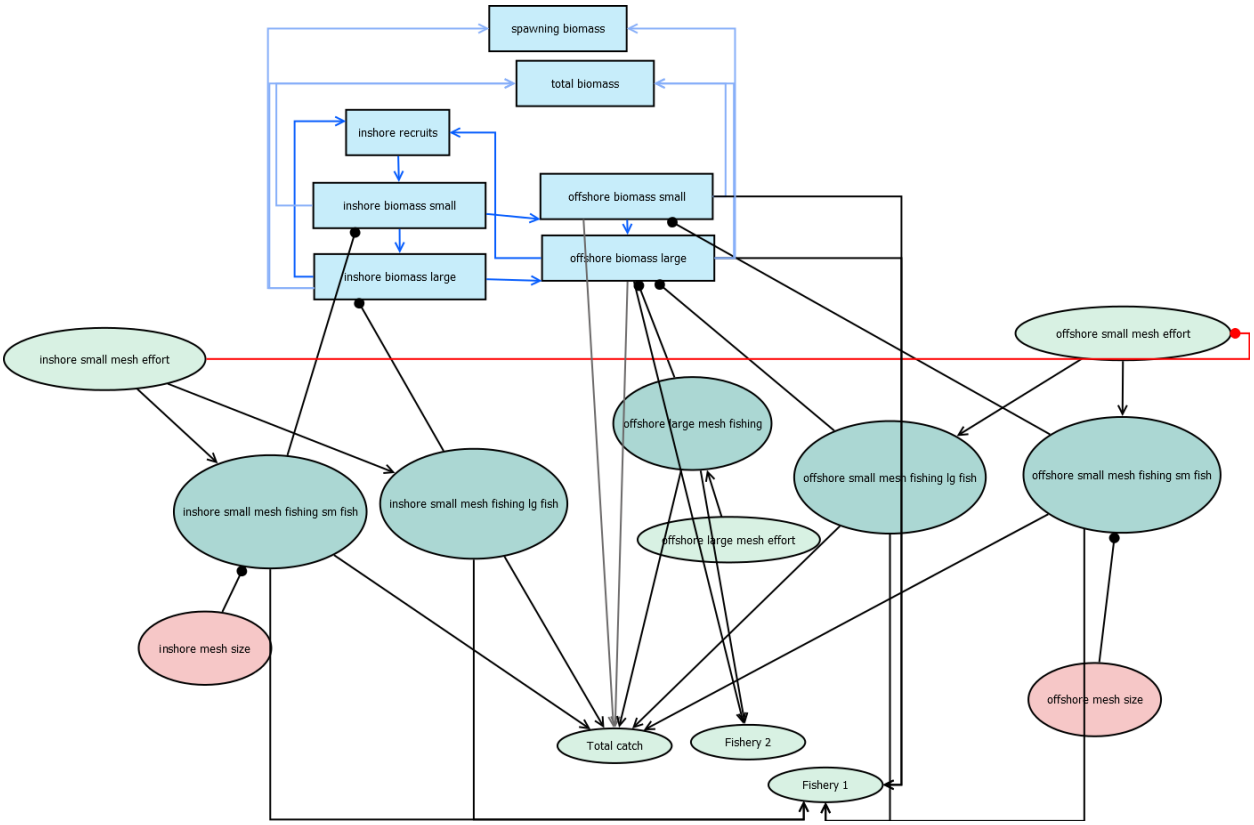


Fig B.2. Management strategy C implemented in the baseline operating model scenario (QNM 0). Incoming links to each performance metric are the same as in Figure 4 in the main text.

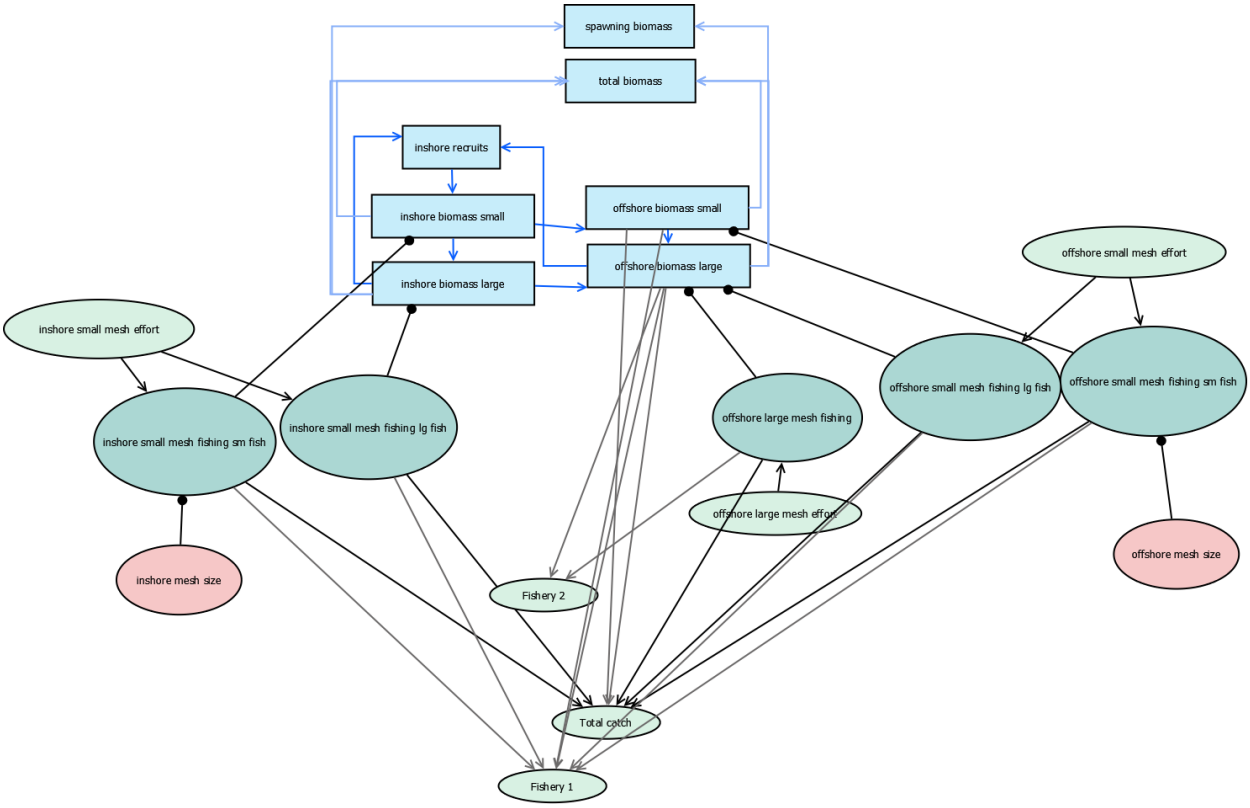


Fig B.3. Management strategy E implemented in the baseline operating model scenario (QNM 0). Incoming links to each performance metric are the same as in Figure 4 in the main text.

Appendix C. Supplementary Results

C.1. Splitting biomass / catch variables by size or class

The low agreement between the QNM and the qMSE for *total catch* and *fishery 1 catch* appeared to result from the inability of the QNM to capture non-linear relationships like density-dependence and gear saturation. We determined this by splitting the catch metrics by size class (QNM; Table S4) or maturity state (qMSE; Table S5), and then comparing model predictions for all fish (i.e., main text results), for small / immature fish, and for large / mature fish (Fig S7).

For example, management strategy D increases the mesh size of fishery 1, reducing the proportion of small fish biomass that fishery 1 can capture. We found more mismatches between QNMs v. qMSE species 1 when we looked only at *small fish fishery 1 catch* rather than all *fishery 1 catch* (Fig S7). Implementing management strategy D in the QNMs led to two important changes in upstream variables from *small fish fishery 1 catch*: a decline in *inshore/offshore fishery 1 fishing* activity (on smaller size classes), and a related increase in *small fish inshore / offshore biomass*. Increased *small fish inshore / offshore biomass* led the QNM to predict a linear increase in *small fish fishery 1 catch* for most scenarios (Table S4). In the qMSE, the additional *small fish inshore / offshore biomass* that survived and grew to a catchable size (still within the "small fish" size class) was limited by density-dependent processes, and the yield of fishery 1 fishing activity on this additional biomass was limited by gear saturation. Consequently, in the qMSE, the benefits to the fishery from an increase in the biomass of small fish did not always outweigh the loss of catch of the smallest fish. We observed similar dynamics in the QNMs when we explored the same increase in mismatches for *small fish total / fishery 1 catch* relative to all *total / fishery 1 catch* for management strategies C & E, and an increase in mismatches for *large fish total / fishery 1 catch* relative to all *total / fishery 1 catch* for management strategy B (Fig S7).

C.2. Incorporating equivocal QNM predictions into a triage process

The QNM output was equivocal (low sign consistency < 60%) in 9-35% of QNMs, depending on the performance metric (Table S2). Recall that the same QNMs were compared across the different species life histories (species 1 v. species 2; Ricker v. Beverton-Holt stock-recruitment relationship).

The response direction of the equivocal QNM outcome was the same as the qMSE prediction about half the time: 52% for species 1, and 48% for species 2 (regardless of stock-recruit model; Table S8). *Total catch* had the greatest number of equivocal responses in the QNMs, while *spawning biomass* only had two equivocal outcomes.

We considered two ways in which equivocal QNM responses could inform a triage process.

The first would be to incorporate the directional response, despite its low certainty, into triage decision-making. When we did this, we saw proportionally fewer false positive outcomes for catch metrics. This means triage would have eliminated more non-viable strategies, which would save time when running the full qMSE. However, equivocal QNM output also included the only false negatives in our study, which would cause the improper elimination of viable strategies

and undercut the purpose of the triage process. We observed several cases for which the equivocal QNM predicted a win-lose biomass-catch tradeoff that the MSE predicted to be a win-win. The question of whether to include equivocal predictions of directional response is therefore a decision of whether it is more important during triage to maximize the elimination of non-viable strategies, or the retention of viable strategies. The latter is likely to be preferable, which would mean conservatively removing equivocal QNM output from triage, as we did for our results.

The second way in which equivocal outcomes could inform a triage process would be to interpret those outcomes as representing no change in the given performance metric. We might then consider a "win" from the QNM to be "positive or no change" in catch or biomass. When we re-evaluated our QNM results in Figure 7 assuming that positive or no change in catch and biomass constitutes a win-win outcome, the QNMs were equally or more optimistic in predicting win-win outcomes compared to the species 2 qMSE (only two out of 12 biomass-catch trade-offs predicted as "no change" win-wins by the QNMs were win-win in the qMSE). For species 1, the QNMs were equally optimistic in predicting win-win outcomes when a "positive or no change" win. Thus the addition of equivocal results in this manner would not have greatly improved QNM performance for triage.

C.3. Figures and Tables

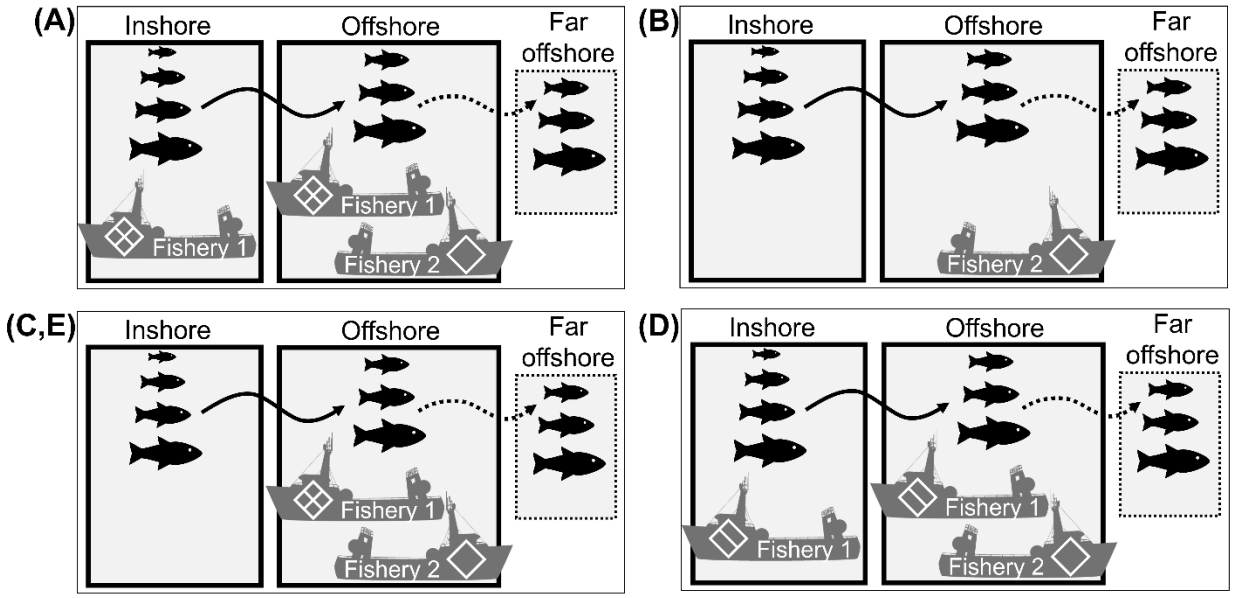


Figure S1. An illustration of the five management strategies A – E. The A. Status Quo strategy is the same as Figure 1. Strategies C and E differ in that for strategy C, fishery 1 effort changes in the offshore to compensate for loss of inshore fishing activity, whereas for strategy E, fishery 1 effort in the offshore is unaffected.

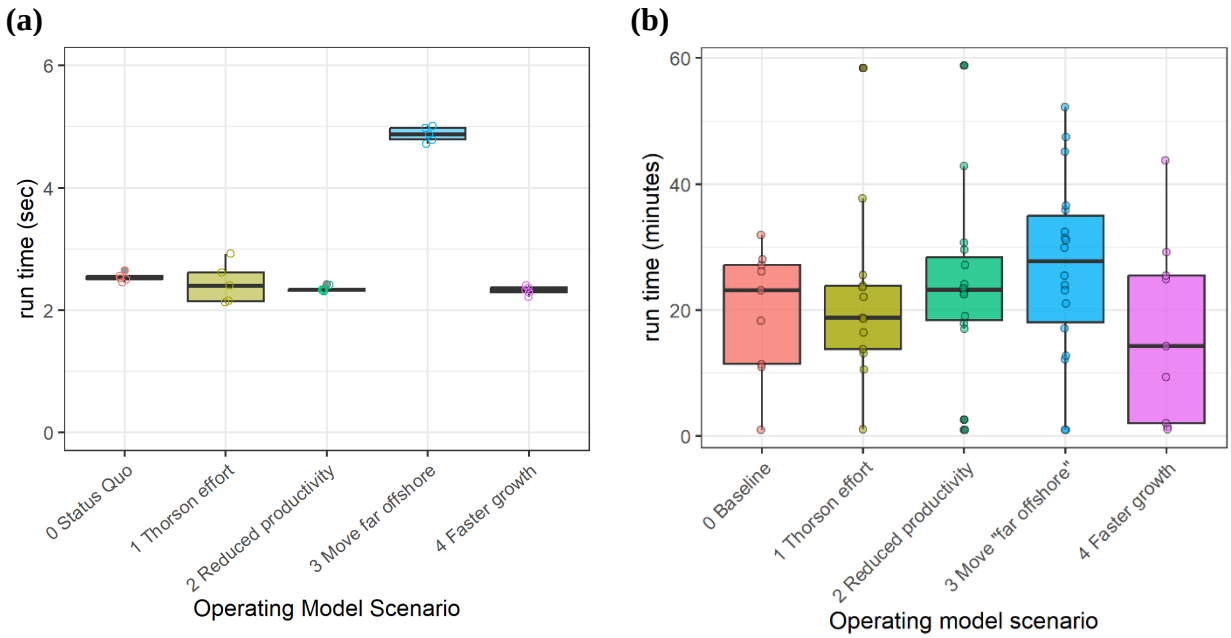


Figure S2. Run time for (a) perturbations implemented with R package *QPress*, by operating model scenario, and (b) boosted regression trees (BRTs) used for sensitivity analysis runs using the R package *gbm*.

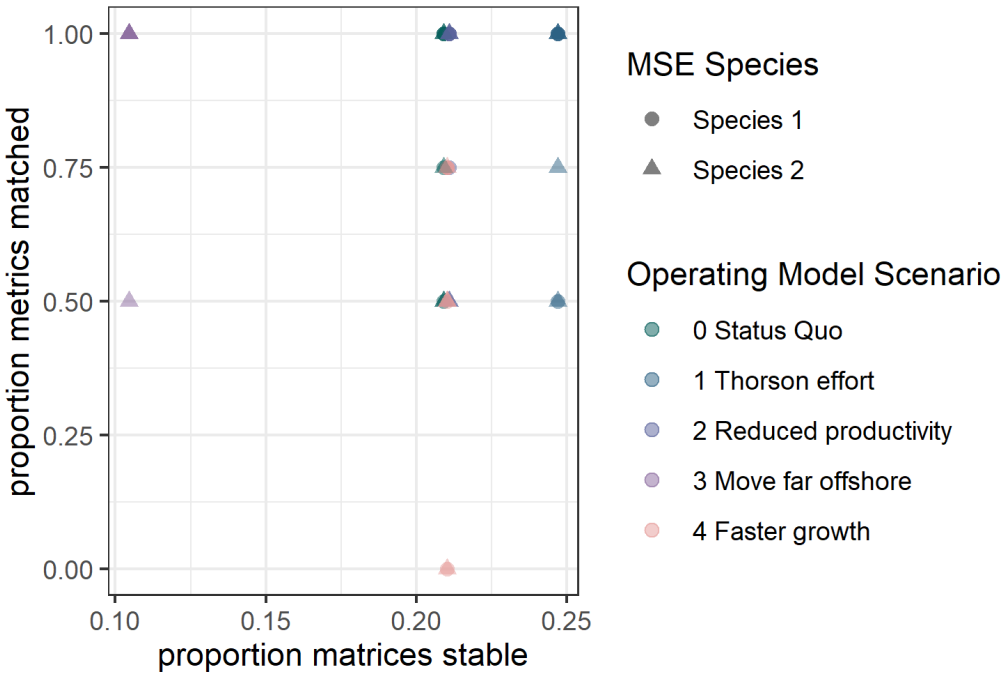


Figure S3. Proportion of stable matrices by operating model, against the proportion (out of 4) of output metrics that matched between the qualitative and quantitative models.

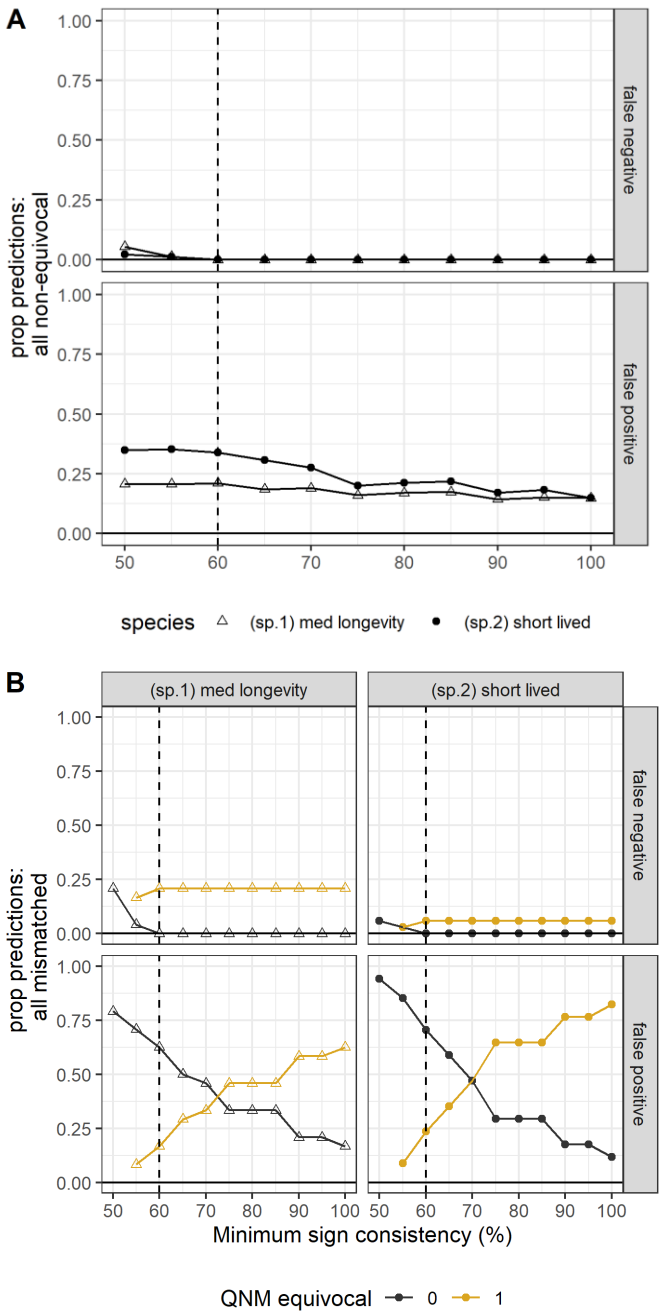


Figure S4. The impact of the cutoff value used to characterize equivocal outcomes ("minimum sign consistency") on the number of false positives / false negatives. The x axis shows the minimum sign consistency used to identify non-equivocal results (such that non-equivocal results must have a sign consistency greater than or equal to x). (A) shows the proportion of false positive or false negative predictions out of all non-equivocal predictions, for each species. (B) shows the proportion of false positive or false negative predictions that are non-equivocal (black series) versus equivocal (yellow series), for each species. Note that at a sign consistency of 51%, no predictions are equivocal. QNMs were compared against the qMSE run with the Ricker stock-recruit model.

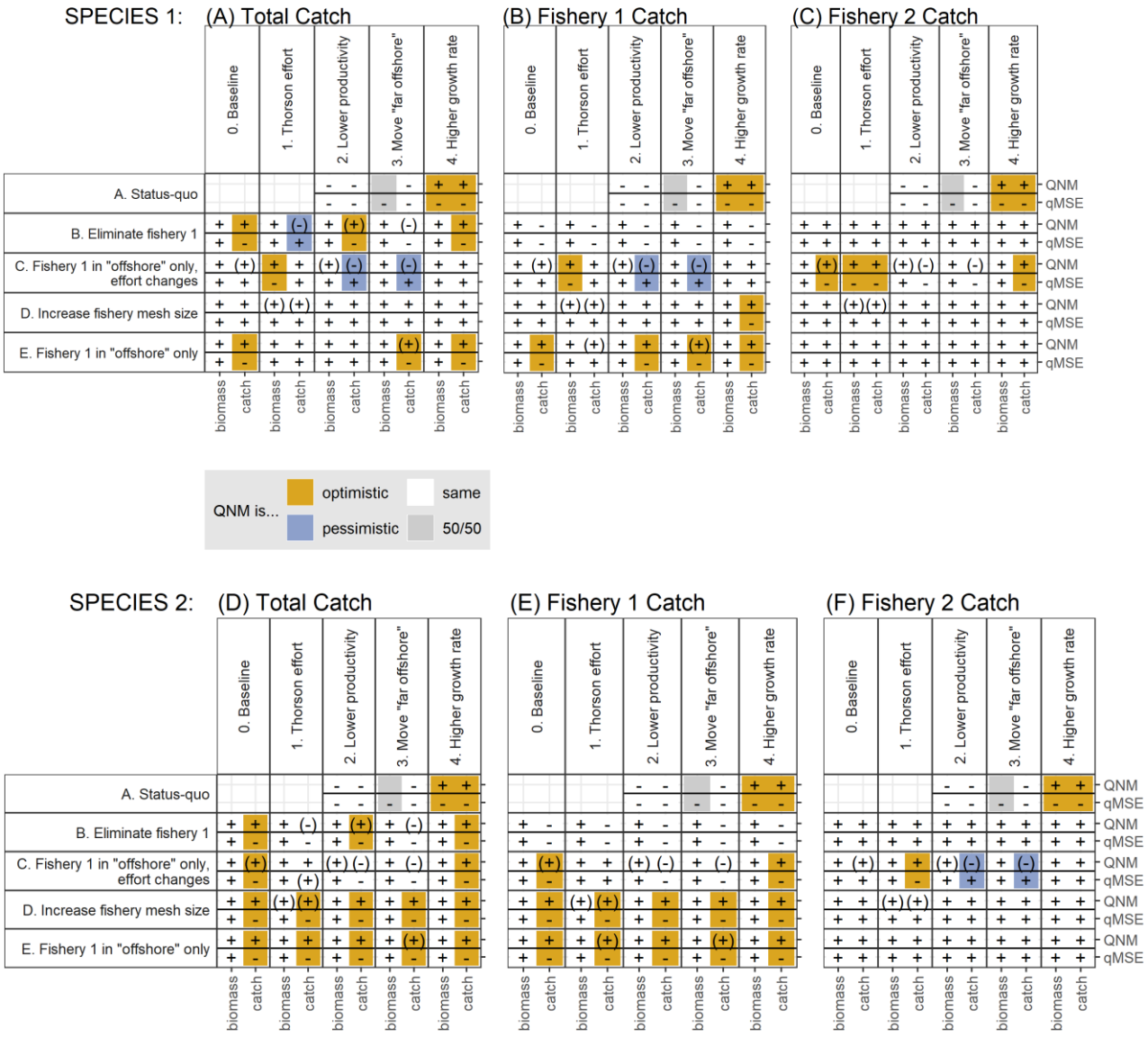


Figure S5. The trade-offs between spawning biomass and (a/d) total catch, (b/e) fishery 1 catch, and (c/f) fishery 2 catch, for each of 23 cases (operating model scenario in column x management strategy in row). Increases are indicated as "+" and decreases as "-," with equivalent outcomes marked with brackets "()." Biomass and catch are reported sequentially (from left to right), with the qualitative model output on the top ("QNM," vertical right axis) and the qMSE output on the bottom ("qMSE," vertical right axis). qMSE output can represent Species 1 (a-c) or Species 2 (d-f), both run with the Ricker stock-recruit model. Shading indicates whether the qualitative model output is more optimistic (yellow) or pessimistic (blue) compared to the quantitative model output, or if the trade-off is the same between the two models (white). For example, for the Status Quo management strategy under operating model scenario 5 (top right corner of each plot) the

qualitative model prediction of the biomass-catch trade-off is more optimistic for all three catch metrics (and therefore shaded gold) for both species; the QNM output reports a win-win (+ catch, + biomass) whereas the full quantitative MSE output reports a lose-lose (- catch, - biomass).

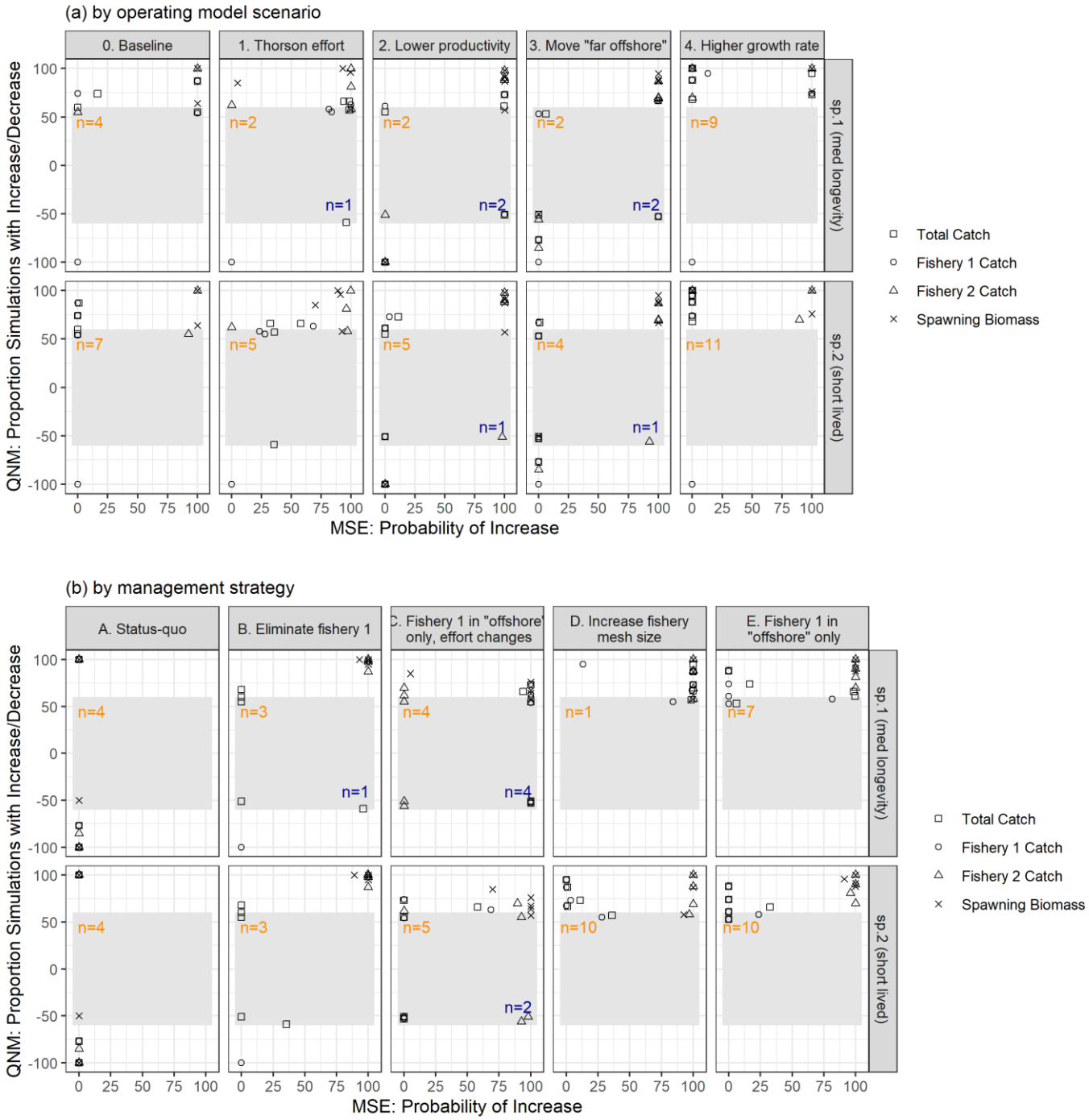


Figure S6. A more quantitative summary of Fig 6 in the main text. The probability that a given variable increased/decreased in the MSE (x-axis), against the proportion of simulations in which the variable increased/decreased in the QNM (y-axis). Increases are indicated by positive x/y axis values, whereas decreases are indicated by negative values. Summarized by (a) operating model scenario, across strategies, and (b) management strategy, across scenarios. MSE probabilities are

relative to the Status Quo management strategy within each scenario (e.g., the probability that Total Catch increased compared to the status-quo when Fishery 1 was eliminated (Strategy B) under the Baseline scenario is represented by a single point in the first column above). Gray area differentiates where the QNM response was equivocal (<60% and >-60% sign consistency). Where more than one point occurred in the upper left / lower right corners (mismatches in response), the total number of points is reported in the graph area (as 'n').

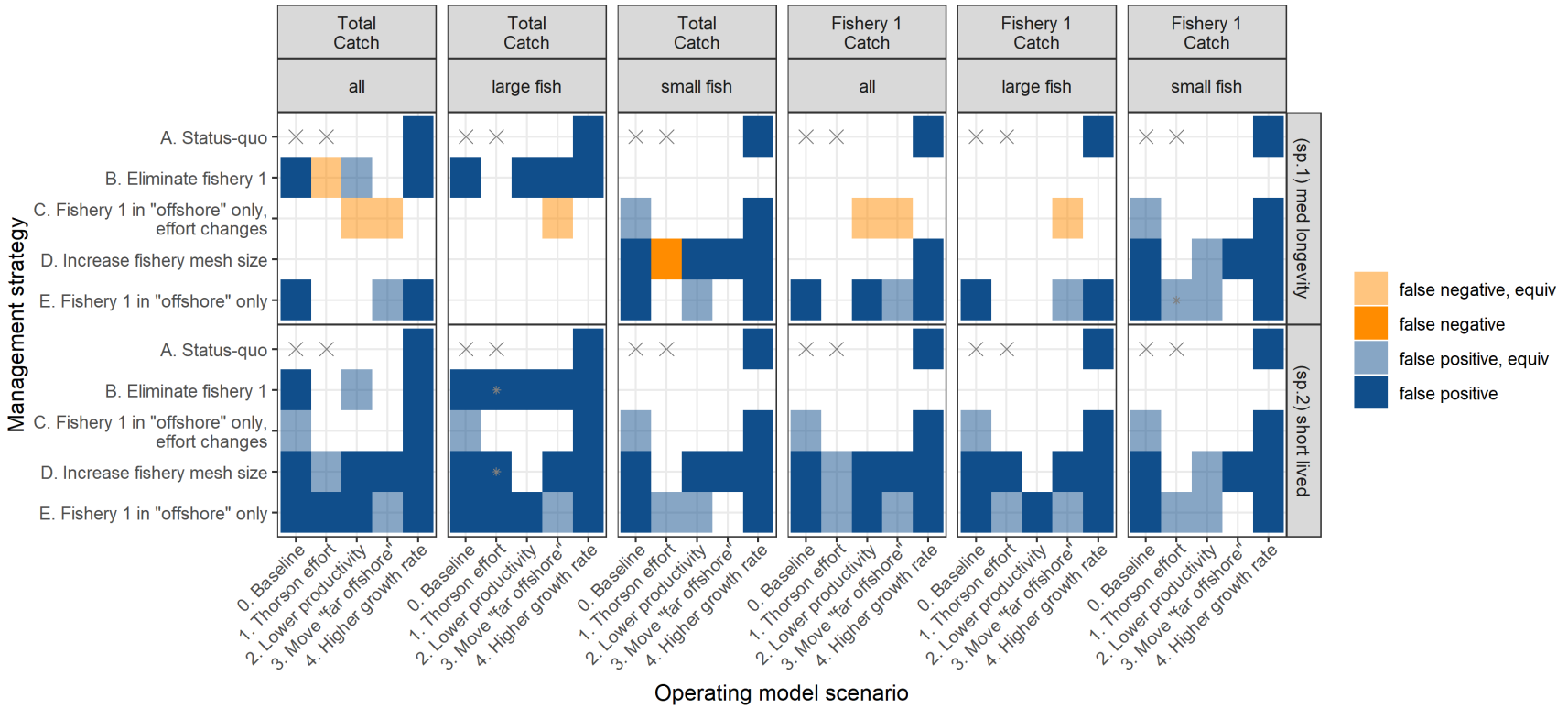


Figure S7. Whether the QNM result was a 'false positive' (orange) or a 'false negative' (blue) when the directional response of Total catch and Fishery 1 catch (columns) did **not** match between the qMSE and the QNM (as in Fig 6). For each performance metric, catch of all fish size classes ("all;" as in main text) is broken down into catch of small fish and catch of large fish. Results are shown for each management strategy (vertical axis) applied under each operating model scenario (horizontal axis). Lighter shades of orange and blue indicate that the QNM result was equivocal ("equiv;" sign consistency of < 60%). Strategy A Status-quo could not be compared between QNMs and qMSE for scenarios 0 and 1, and so are marked by an "X." Fig S4 demonstrates the effect of the equivocal sign consistency cut-off on false negative / false positive classifications.

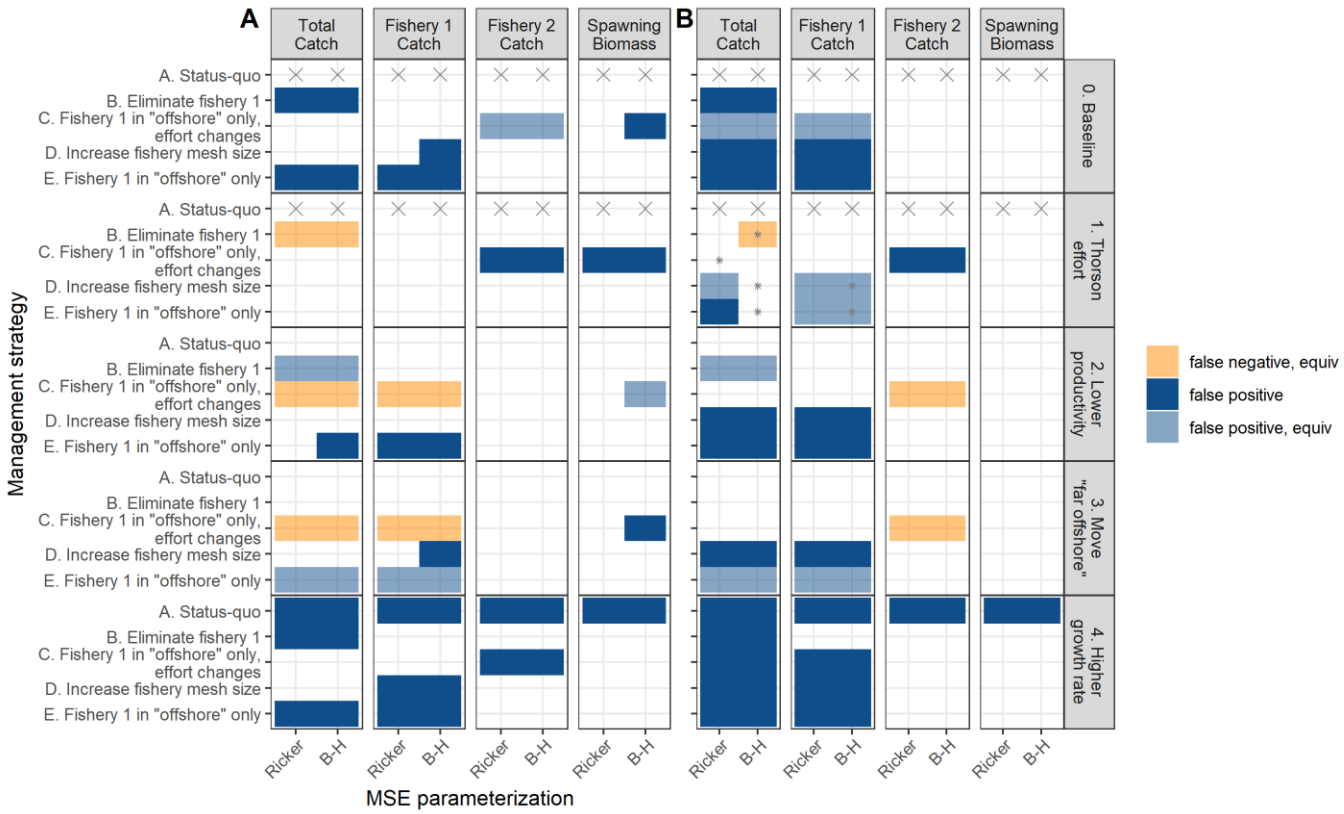


Figure S8. Whether the QNM result was a 'false positive' (orange) or a 'false negative' (blue) when the directional response of Total catch and Fishery 1 catch (columns) did **not** match between the qMSE and the QNM (as in Fig 6). The horizontal axis indicates whether the qMSE used the Ricker stock-recruit model (as in the main text), or the Beverton-Holt model ("B-H recruit"). Results are shown for each management strategy (vertical axis) applied under each operating model scenario (rows). Lighter shades of orange and blue indicate that the QNM result was equivocal ("equiv;" sign consistency of < 60%). Strategy A Status-quo could not be compared between QNMs and qMSE for scenarios 0 and 1, and so are marked by an "X." Fig S4 demonstrates the effect of the equivocal sign consistency cut-off on false negative / false positive classifications. Equivocal qMSE output is indicated with a "*".

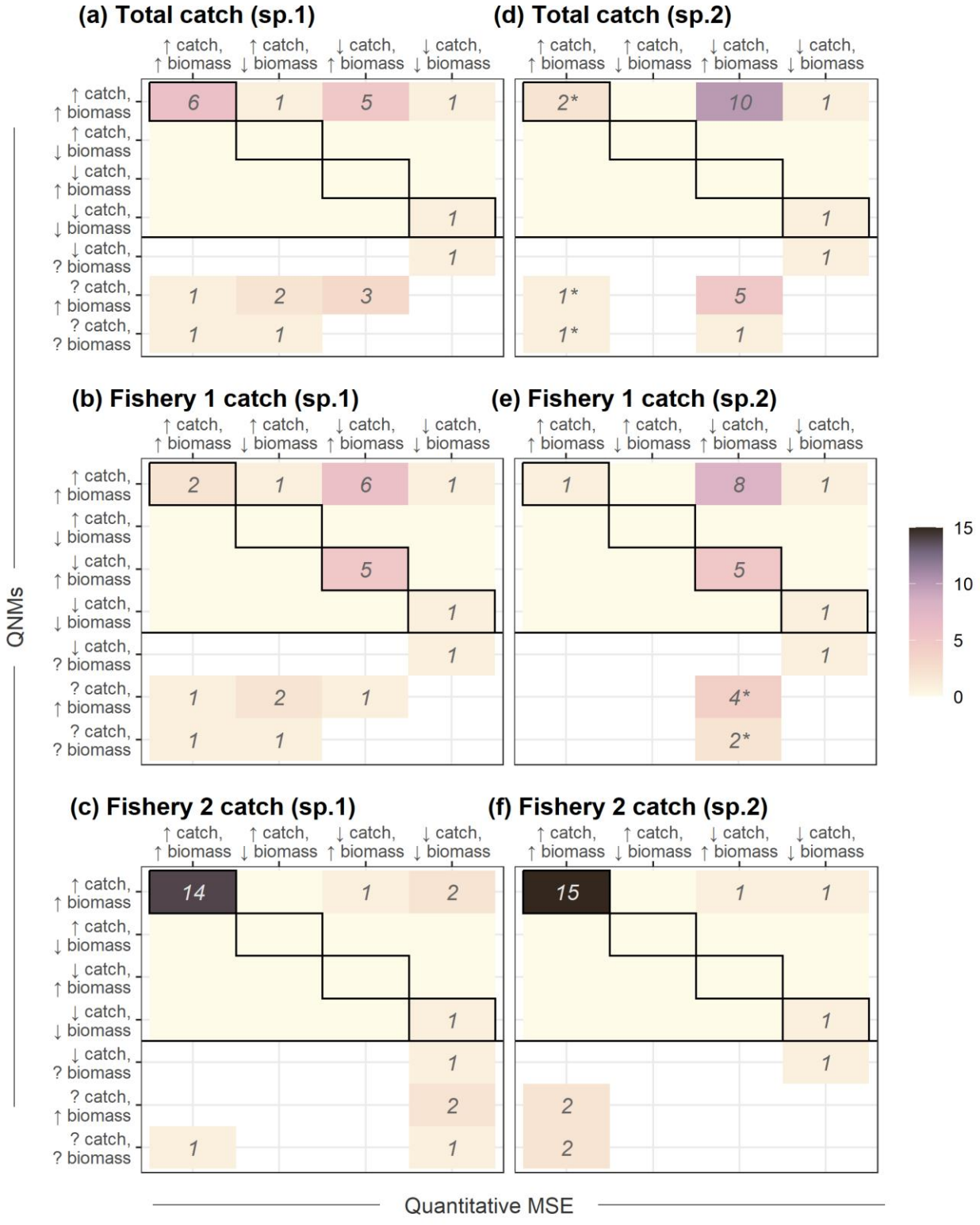
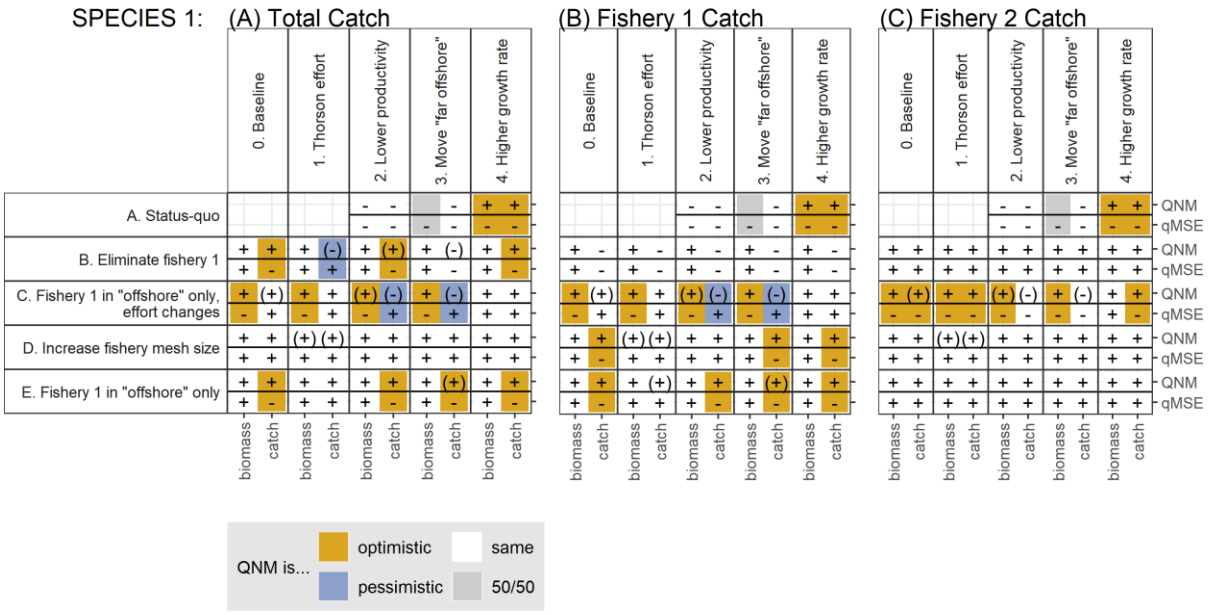


Figure S9. As in Fig 7 in the main text, but when qMSEs were run using the Beverton-Holt stock-recruit relationship. Trade-offs for spawning biomass and (a,d) total catch, (b,e) fishery 1 catch, and (c,f) fishery 2 catch, predicted by the qualitative (vertical axis) and the quantitative (horizontal axis) models. '↑' indicates a predicted increase and '↓' a decrease; for both qualitative and

quantitative models, outcomes can be a “win-win” (↑ catch, ↑ biomass; top row and left column) a “lose-lose” (↓ catch, ↓ biomass; bottom row and right column), or a trade-off (↓ catch, ↑ biomass; ↑ catch, ↓ biomass). Cells on the diagonal (outlined in black) are cases when both the qualitative and quantitative models predicted the same direction of change for catch and biomass. Model outcomes can also be considered equivocal. Counts that include qMSE outcomes that were equivocal are annotated with a (*). Equivocal QNM outcomes are included on the vertical axis below a horizontal line, and written as ‘?.’ The cell fill color indicates the number of operating model scenario x management strategy cases which match the cell conditions (out of 23 possible cases).



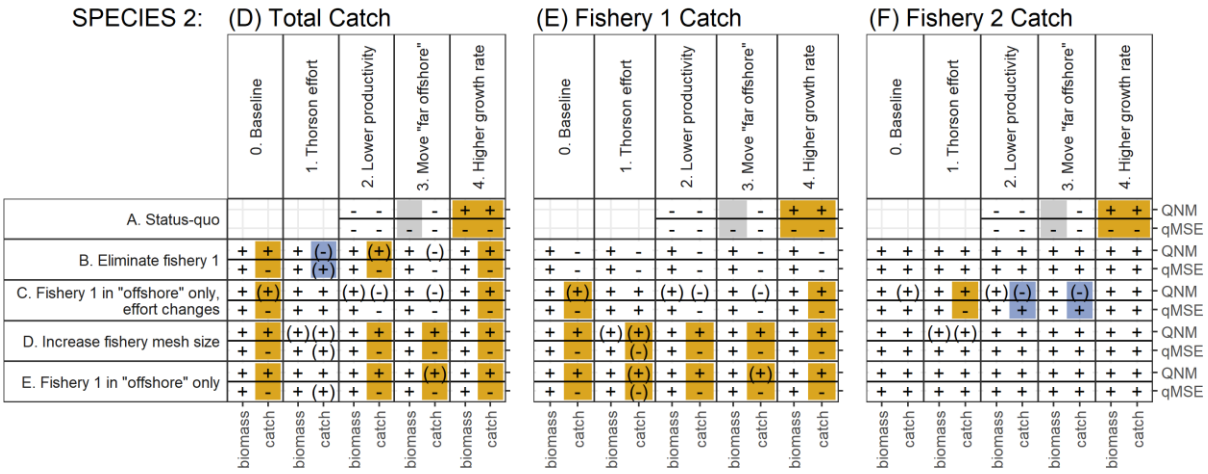


Figure S10. As in Fig S5, but for qMSE run with the Beverton-Holt model stock-recruit model. The trade-offs between spawning biomass and (a/d) total catch, (b/e) fishery 1 catch, and (c/f) fishery 2 catch, for each of 23 cases (operating model scenario in column x management strategy in row). Increases are indicated as "+" and decreases as "-," with equivalent outcomes marked with brackets "()." Biomass and catch are reported sequentially (from left to right), with the qualitative model output on the top ("QNM," vertical right axis) and the qMSE output on the bottom ("qMSE," vertical right axis). qMSE output can represent Species 1 (a-c) or Species 2 (d-f), both run with the Ricker stock-recruit model. Shading indicates whether the qualitative model output is more optimistic (yellow) or pessimistic (blue) compared to the quantitative model output, or if the trade-off is the same between the two models (white). For example, for the Status Quo management strategy under operating model scenario 5 (top right corner of each plot) the

qualitative model prediction of the biomass-catch trade-off is more optimistic for all three catch metrics (and therefore shaded gold) for both species; the QNM output reports a win-win (+ catch, + biomass) whereas the full quantitative MSE output reports a lose-lose (- catch, - biomass).

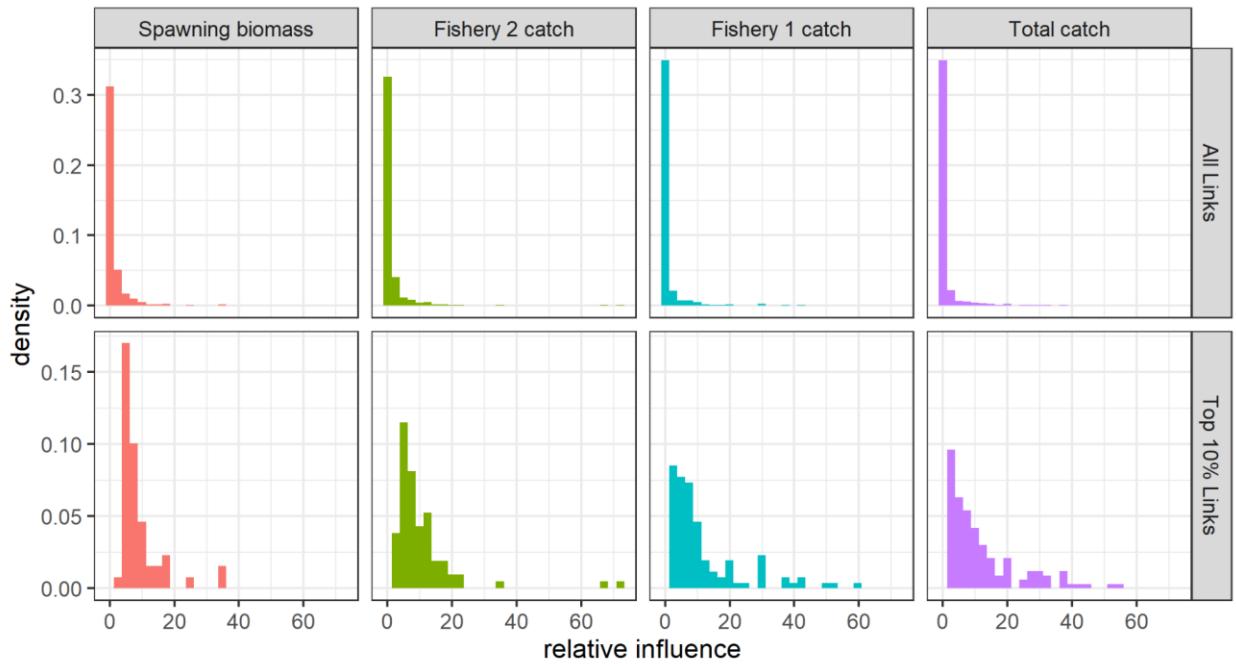


Figure S11. The distribution of relative influence estimates from boosted regression trees. Results are compiled across all operating model scenario x management strategy pairs for each variable. Variables are ordered according to the overall degree of agreement between the qualitative v. quantitative models, from right (most agreement) to left (least agreement). The lower panels show only the highest 10% of relative influence values for each variable, to better visualize the shape of each distribution.

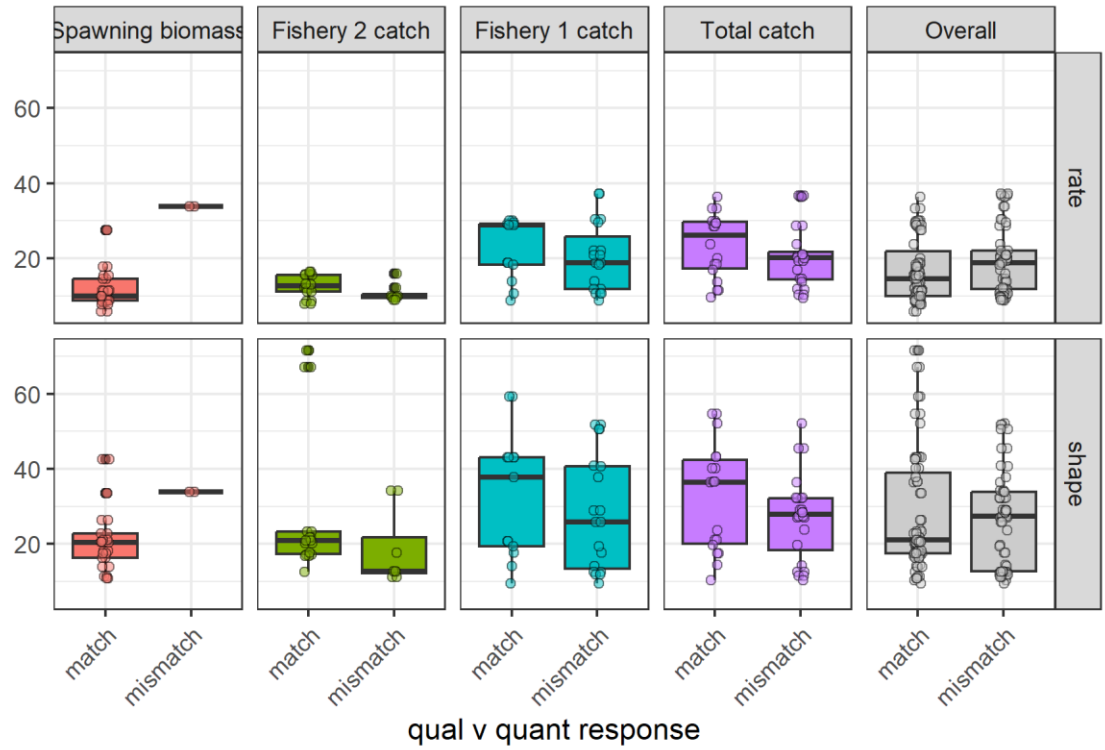


Figure S12. The shape and rate parameters for the distribution of link-specific relative influences (from boosted regression trees), for which the response direction of the given output metric (columns) from the qualitative models either matched or mismatched (x axis) the response direction from the quantitative models (run with the Ricker stock-recruit model).

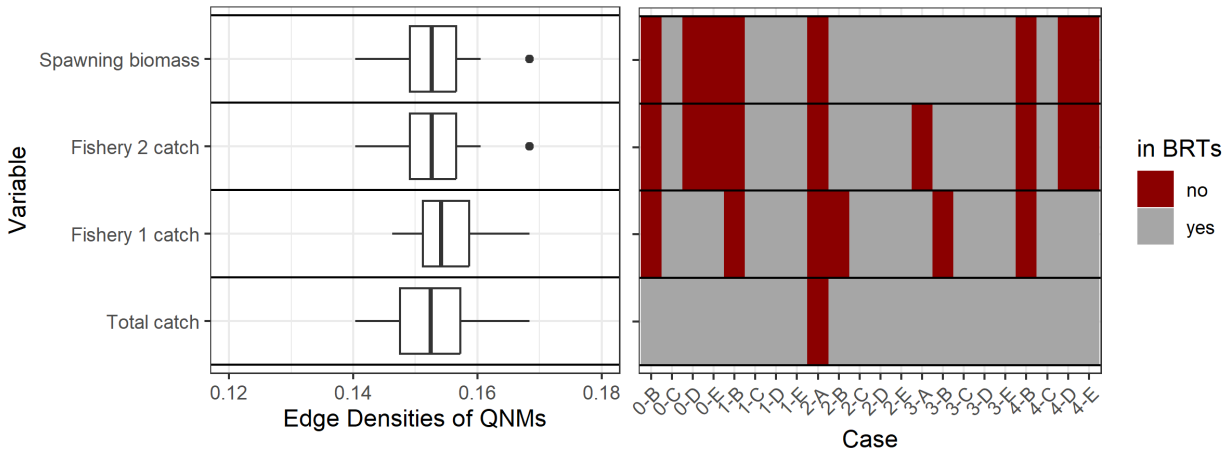


Figure S13. The (a) distribution of edge densities of the QNMs which informed the boosted regression trees (BRTs) for each variable. Since not all QNMs informed the BRTs (a metric's response to a perturbation had to vary across 10,000 stable matrices to be informative), graph (b) identifies which QNMs, out of all operating model x management strategy pairs (horizontal axis), did not (red) inform the BRTs for each variable (vertical axis). For example, *Total catch* response

always varied in response across 10,000 stable matrices, so all QNMs informed the BRTs, whereas *Spawning Biomass* only varied in response in 8 / 20 QNMs, so 12 QNMs (identified with red shading) did not inform the BRTs.

Table S1. Full results from the qMSE for (a) species 1 and (b) species 2, comparing the main text results using the Ricker ("Ricker") with the Beverton-Holt stock-recruit relationship (B-H). Bolded cell borders indicate changes in predicted response direction for the given performance metric / case.

Results are reported as the probability that each output metric increases relative to management strategy A (strategies B-E), or case 0-A (strategy A). The color of the cell indicates whether there was a non-equivocal predicted increase (green) or decrease (orange). A probability > 50% was considered a predicted increase, and a probability <50% was considered a predicted decrease. For comparison to the QNMs, probabilities 40% < x < 60% were classified as equivocal, and are italicized in uncolored cells.

(a)

Species 1

	Catch							
	Total		Fishery 1		Fishery 2		Spawning biomass	
Case	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
0-B	0	0	0	0	100	100	100	100
0-C	100	100	100	100	0	0	100	35
0-D	100	100	100	11	100	100	100	100
0-E	16.5	0	0	0	100	100	100	100
1-A	0	0	0	0	0	0	0	0
1-B	96	98	0	0	100	100	93	75.5
1-C	94	98.5	100	100	0	0	5	1
1-D	98.5	99	84	83	100	100	99.5	100
1-E	98.5	98	81.5	85.5	100	100	99.5	99.5
2-A	0	0	0	0	0	0	0	0
2-B	0	0	0	0	100	100	100	100
2-C	100	100	100	100	0	0	100	0
2-D	100	100	100	100	100	100	100	100
2-E	99.5	8.5	0	0	100	100	100	100
3-A	0	0	0	0	0	0	0	0
3-B	0	0	0	0	100	100	100	100
3-C	100	100	100	100	0	0	100	36.5
3-D	100	100	99	6.5	100	100	100	100
3-E	6	0	0	0	100	100	100	100
4-A	0	0	0	0	0	0	0	0
4-B	0	0	0	0	100	100	100	100
4-C	100	100	100	100	0	0	100	97.5
4-D	100	66.5	13	0	100	100	100	100
4-E	0	0	0	0	100	100	100	100

(b)

Species 2								
	Catch						Spawning biomass	
	Total		Fishery 1		Fishery 2			
Case	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
0-B	0	0	0	0	100	100	100	100
0-C	0	0	0	0	92.5	100	100	100
0-D	1	0.5	0	0	100	100	100	100
0-E	0	0	0	0	100	100	100	100
1-A	0	0	0	0	0	0	0	0
1-B	35.5	*57.5*	0	0	99.5	100	89	99
1-C	58	74	*68.5*	78.5	0	0	70	67.5
1-D	36	*58.5*	28	*48*	97	100	92.5	100
1-E	32.5	*57*	23.5	*47.5*	96	100	91	99.5
2-A	0	0	0	0	0	0	0	0
2-B	0	0	0	0	100	100	100	100
2-C	0	0	0	0	98	100	100	100
2-D	11	1.5	3.5	0.5	100	100	100	100
2-E	0	0	0	0	100	100	100	100
3-A	0	0	0	0	0	0	0	0
3-B	0	0	0	0	100	100	100	100
3-C	0	0	0	0	92.5	100	100	100
3-D	1	0.5	0	0	100	100	100	100
3-E	0	0	0	0	100	100	100	100
4-A	0	0	0	0	0	0	0	0
4-B	0	0	0	0	100	100	100	100
4-C	0	0	0	0	89.5	100	100	100
4-D	0	0	0	0	100	100	100	100
4-E	0	0	0	0	100	100	100	100

Table S2. The number of cases (i.e., operating model scenario x management strategy pairs) out of 23 which had an equivocal, non-equivocal, or uninformative QNM prediction for each performance metric. Equivocal outcomes are those for which we have low certainty in a variable's response direction because <60% of the perturbed stable matrices predict the same response direction. Uninformative outcomes are those for which exactly 50% of the perturbed stable matrices predict the same response direction.

	N (equivocal)	**N (un-equivocal)**	**N (uninformative)**
Spawning Biomass	2	20	1
Fishery 2 catch	4	19	0
Fishery 1 catch	6	17	0
Fishery 1 catch small fish	7	16	0
Fishery 1 catch large fish	5	18	0
Total catch	8	15	0
Total catch small fish	6	17	0
Total catch large fish	4	19	0

Table S3. Matches and mismatches in outcomes between the qMSE and the corresponding QNMs, when qMSE used the Ricker stock-recruit model ("Ricker") v the Beverton-Holt stock-recruit model ("B-H"). (Mis)matches are quantified as the proportion of comparable cases (out of *N*) for which the predicted direction of change for the given metric was the same ("match") or different ("mismatch"). The top half of the table provides the proportion of all QNM predictions (*N)* that unequivocally matched / mismatched the qMSE (≥ 60% sign consistency), or were equivocal (50% < x < 60% sign consistency) or uninformative (50% sign consistency). The lower half of the table provides the proportion of out of only non-equivocal QNM predictions (*N*) that matched / mismatched the qMSE. The last row of the table lists the operating model x management strategy cases for which there were directional mismatches. Values are shown for the Beverton-Holt model qMSE only when they differed from the Ricker model qMSE. Proportions that include qMSE outcomes that were equivocal are annotated with a ($^{+}$), with the number of ($^{+}$) indicating the number of equivocal outcomes.

(A) Species 1: medium-longevity

	Catch						Spawning biomass		Overall	
	Total		Fishery 1		Fishery 2					
	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
All QNMs										
N	*23*	*23*	*23*	*23*	*23*	*23*	*23*	*23*	*92*	*92*
Match:	0.43	0.39	0.52	0.43	0.70		0.78	0.70	0.61	0.55
Mismatch:	0.22	0.26	0.22	0.30	0.13		0.09	0.17	0.16	0.22
Equivocal:	0.35	0.35	0.26		0.17		0.09		0.22	
Uninformative:							0.04	0.04	0.01	
Only non-equivocal QNMs										
N	*15*	*15*	*17*	*17*	*19*	*19*	*20*	*20*	*71*	*71*
Match:	0.80	0.60	0.53	0.59	0.89	0.84	0.90	0.80	0.79	0.72
Mismatch:	0.20	0.40	0.47	0.41	0.11	0.16	0.10	0.20	0.21	0.28

(B) Species 2: short-lived

	Catch						Spawning biomass		Overall	
	Total		Fishery 1		Fishery 2					
	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
All QNMs										
N	*23*	*23*	*23*	*23*	*23*	*23*	*23*	*23*	*92*	*92*
Match:	0.13+	0.17+	0.35		0.74		0.83		0.51	0.52
Mismatch:	0.52	0.48	0.39		0.09		0.04		0.26	0.25
Equivocal:	0.34	0.34++	0.26	0.26++	0.18		0.09		0.22	
Uninformative:							0.04		0.01	
Only non-equivocal QNMs										
N	*15*	*15*	*17*	*17*	*19*	*19*	*20*	*20*	*71*	*71*
Match:	0.67+	0.27+	0.53	0.47	0.89		0.95		0.67	0.68
Mismatch:	0.33	0.73	0.47	0.53	0.11		0.05		0.33	0.32

Table S4. QNM results when t*otal catch* and *fishery 1 catch* were broken down by small and large fish. Results are reported as the percent of 10,000 stable matrices in which each output metric increased / decreased relative to the unperturbed state. Percentages are only reported for the direction of change exhibited by the majority of matrices; the sign of the percentage and color of the cell indicates whether that majority direction was an increase (+, green) or decrease (-, orange). Equivocal outcomes (50% < x < 60% stable matrices with the same response direction) are uncolored and italicized.

Case	Operating model	Management strategy	Total Catch			Fishery 1 Catch		
			All	Small fish	Large fish	All	Small fish	Large fish
0-B	0. Baseline	B. Eliminate fishery 1	60	-100	77	-100	-100	-100
0-C	0. Baseline	C. Fishery 1 in "offshore" only, effort spillover	*55*	52	56	54	52	56
0-D	0. Baseline	D. Increase fishery mesh size	87	73	100	87	73	100
0-E	0. Baseline	E. Fishery 1 in "offshore" only	74	63	76	74	64	76
1-B	1. Thorson effort	B. Eliminate fishery 1	*-59*	-100	60	-100	-100	-100
1-C	1. Thorson effort	C. Fishery 1 in "offshore" only, effort spillover	66	60	66	63	60	61
1-D	1. Thorson effort	D. Increase fishery mesh size	*57*	-82	94	55	-82	99
1-E	1. Thorson effort	E. Fishery 1 in "offshore" only	66	55	70	58	55	57
2-A	2. Lower productivity	A. Status-quo	-100	-100	-100	-100	-100	-100
2-B	2. Lower productivity	B. Eliminate fishery 1	*55*	-100	73	-100	-100	-100
2-C	2. Lower productivity	C. Fishery 1 in "offshore" only, effort spillover	*-51*	-53	-50	-51	-54	-50
2-D	2. Lower productivity	D. Increase fishery mesh size	73	60	87	73	59	87
2-E	2. Lower productivity	E. Fishery 1 in "offshore" only	61	51	66	61	51	66
3-A	3. Move "far offshore"	A. Status-quo	-77	-72	-79	-77	-72	-79
3-B	3. Move "far offshore"	B. Eliminate fishery 1	*-51*	-100	64	-100	-100	-100
3-C	3. Move "far offshore"	C. Fishery 1 in "offshore" only, effort spillover	*-53*	-54	-54	-53	-55	-54
3-D	3. Move "far offshore"	D. Increase fishery mesh size	67	60	76	67	61	75
3-E	3. Move "far offshore"	E. Fishery 1 in "offshore" only	*53*	-53	55	53	-53	55
4-A	4. Higher growth rate	A. Status-quo	100	100	100	100	100	100
4-B	4. Higher growth rate	B. Eliminate fishery 1	68	-100	82	-100	-100	-100
4-C	4. Higher growth rate	C. Fishery 1 in "offshore" only, effort spillover	73	72	71	74	72	70
4-D	4. Higher growth rate	D. Increase fishery mesh size	95	87	100	95	87	100
4-E	4. Higher growth rate	E. Fishery 1 in "offshore" only	88	81	86	88	81	86

Table S5. qMSE results when *total catch* and *fishery 1 catch* were broken down by juvenile and adult fish. Results are reported as the probability that each output metric increases relative to management strategy A (Status Quo) for the given operating model scenario (strategies B-E), or, for strategy A, case 0-A. The color of the cell indicates whether there was a non-equivocal predicted increase (green) or decrease (orange). A probability > 50% was considered a predicted increase, and a probability <50% was considered a predicted decrease. For comparison to the QNMs, probabilities 40% < x < 60% were classified as equivocal, and are italicized in uncolored cells. The qMSE used the Ricker stock-recruit model.

	Species 1						Species 2					
	Total catch			Fishery 1 catch			Total catch			Fishery 1 catch		
Case	All	Juvenile fish	Adult fish	All	Juvenile fish	Adult fish	All	Juvenile fish	Adult fish	All	Juvenile fish	Adult fish
0-B	0	0	0	0	0	0	0	0	0	0	0	0
0-C	100	0	100	100	0	100	0	0	0	0	0	0
0-D	100	0	100	100	0	100	1	0	38	0	0	12
0-E	16.5	0	100	0	0	4.5	0	0	0	0	0	0
1-B	96	40	98.5	0	0	0	35.5	19.5	41	0	0	0
1-C	94	96	81.5	100	100	100	58	52	61	68.5	55.5	73
1-D	98.5	71.5	100	84	45	98.5	36	24	42	28	19.5	32
1-E	98.5	69	100	81.5	45	97	32.5	22.5	37	23.5	17.5	25.5
2-A	0	0	0	0	0	0	0	0	0	0	0	0
2-B	0	0	0	0	0	0	0	0	0	0	0	0
2-C	100	0	100	100	0	100	0	0	0	0	0	0
2-D	100	0	100	100	0	100	11	0	81.5	3.5	0	60
2-E	99.5	0	100	0	0	93.5	0	0	0	0	0	0
3-A	0	0	0	0	0	0	0	0	0	0	0	0
3-B	0	0	0	0	0	0	0	0	0	0	0	0
3-C	100	0	100	100	0	100	0	0	0	0	0	0
3-D	100	0	100	99	0	100	1	0	36.5	0	0	11
3-E	6	0	100	0	0	1	0	0	0	0	0	0
4-A	0	0	0	0	0	0	0	0	0	0	0	0
4-B	0	0	0	0	0	0	0	0	0	0	0	0
4-C	100	0	100	100	0	100	0	0	0	0	0	0
4-D	100	0	100	13	0	100	0	0	8.5	0	0	1
4-E	0	0	100	0	0	0	0	0	0	0	0	0

Table S6. Information shown in (a) Fig 6 (Ricker model) and (b) Fig S8 (Beverton-Holt model), tabulated. The number of metrics (out of 4) in each operating model scenario (rows) x management strategy (columns) for which the response direction did not match between the quantitative and qualitative models. Cases of directional mismatch where the QNM result was categorized as equivocal are not included in these counts. QNM comparisons to the quantitative MSE Species 1 and Species 2 are shown separately. The maximum possible number of mismatches is n=20 for OM scenarios 2-4, and n=16 for OM scenarios 0-1; the maximum possible number of mismatches is n=20 for strategies B-E, and n=12 for strategy A Status Quo.

(A)

		Management Strategy					
	Operating Model Scenario	A. Status Quo	B. Eliminate Fishery 1	C. Fishery 2 in offshore only, effort spillover	D. Increase mesh size	E. Fishery 1 in offshore only	*total*
Species 1	0. Baseline		1	0	0	2	*3*
	1. Thorson effort		0	2	0	0	*2*
	2. Lower productivity	0	0	0	0	1	*1*
	3. Move far offshore	0	0	0	0	0	*0*
	4. Higher growth rate	4	1	1	1	2	*9*
	total	*4*	*2*	*3*	*1*	*5*	*15*
Species 2	0. Baseline		1	0	2	2	*5*
	1. Thorson effort		0	1	0	1	*2*
	2. Lower productivity	0	0	0	2	2	*4*
	3. Move far offshore	0	0	0	2	0	*2*
	4. Higher growth rate	4	1	2	2	2	*11*
	total	*4*	*2*	*3*	*8*	*7*	*24*

(B)

		Management Strategy					
	Operating Model Scenario	A. Status Quo	B. Eliminate Fishery 1	C. Fishery 2 in offshore only, effort spillover	D. Increase mesh size	E. Fishery 1 in offshore only	*total*
Species 1	0. Baseline		1	1	1	2	*5*
	1. Thorson effort		0	2	0	0	*2*
	2. Lower productivity	0	0	0	0	2	*2*
	3. Move far offshore	0	0	1	1	0	*2*
	4. Higher growth rate	4	1	1	1	2	*9*
	total	*4*	*2*	*5*	*3*	*6*	*20*
Species 2	0. Baseline		1	0	2	2	*5*
	1. Thorson effort		0	1	0	0	*1*
	2. Lower productivity	0	0	0	2	2	*4*
	3. Move far offshore	0	0	0	2	0	*2*
	4. Higher growth rate	4	1	2	2	2	*11*
	total	*4*	*2*	*3*	*8*	*6*	*23*

Table S7. False positive rates by strategy / metric. Rates represent the proportion of operating model scenarios (out of a maximum of 5 for managements strategies B-E, and 3 for management strategy A) which returned a false positive. The total number of operating model scenarios underlying each statistic in (a) may be less than the corresponding statistic in (b) because equivocal and uninformative results are excluded. We show QNM comparisons to the qMSE using the Ricker stock-recruit model ("Ricker") v. the Beverton-Holt stock-recruit model ("B-H"). Values are shown for the Beverton-Holt model qMSE only when they differ from the Ricker model qMSE. Proportions that include qMSE outcomes that were equivocal are annotated with a (+), with the number of (+) indicating the number of equivocal outcomes.

	Species 1: medium longevity							
	Catch						Spawning Biomass	
	Total		Fishery 1		Fishery 2			
	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
A. Status-quo	0.33		0.33		0.33		0.33	
B. Eliminate fishery 1	0.4 (0.6)		0		0		0	
C. Fishery 1 in "offshore" only, effort changes	0		0		0.4 (0.6)		0.2	0.6 (0.8)
D. Increase fishery mesh size	0		0.2	0.6	0		0	
E. Fishery 1 in "offshore" only	0.4 (0.6)	0.6 (0.8)	0.6 (0.8)		0		0	

	Species 2: short-lived							
	Catch						Spawning Biomass	
	Total		Fishery 1		Fishery 2			
	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
A. Status-quo	0.33		0.33		0.33		0.33	
B. Eliminate fishery 1	0.4 (0.6)		0		0		0	
C. Fishery 1 in "offshore" only, effort changes	0.2 (0.4)		0.2 (0.4)		0.2		0	
D. Increase fishery mesh size	0.8 (1)	0.8	0.8 (1)	0.8 (1)$^{+}$	0		0	
E. Fishery 1 in "offshore" only	0.8 (1)	0.6 (0.8)	0.6 (1)	0.6 (1)$^{+}$	0		0	

$^{+}$ Includes 1 case in which the qMSE outcome was equivocal

Table S8. Matches and mismatches in outcomes between the qMSE and the corresponding QNMs when the QNM output was equivocal (50% < x < 60% stable matrices showed the same response direction). Proportions are out of the number of QNMs equivocal (top row). We show QNM comparisons to the qMSE using the Ricker stock-recruit model ("Ricker") v. the Beverton-Holt stock-recruit model ("B-H"). Values are shown for the Beverton-Holt model qMSE only when they differ from the Ricker model qMSE. Proportions that include qMSE outcomes that were equivocal are annotated with a (+), with the number of (+) indicating the number of equivocal outcomes.

(A) Species 1: medium-longevity

	Catch						Spawning Biomass		Overall	
	Total		Fishery 1		Fishery 2					
	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
QNMs equivocal (n)	8		4		4		3		21	
Match (proportion)	0.375	0.375	0.5		0.75		0.67	0.33	0.52	0.524
Mismatch (proportion)	0.625		0.5		0.25		0	0.33	0.45	0.428
False positive	0.250		0.17		0.25		0	0.33	0.20	0.190
False negative	0.375	0.375	0.33		0		0	0	0.25	0.238
QNM split (proportion)	0		0		0		0.33	0.33	0.25	0.048

(B) Species 2: short-lived

	Catch						Spawning Biomass		Overall	
	Total		Fishery 1		Fishery 2					
	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H	Ricker	B-H
QNMs equivocal (n)	8		6		4		3		21	
Match (proportion)	0.50	0.500^{+}	0.33	0.33^{+}	0.5		0.67		0.476	
Mismatch (proportion)	0.50	0.500	0.67		0.5		0		0.476	
False positive	0.50	0.375	0.67	0.67^{++}	0		0		0.381	
False negative	0	0.125^{+}	0		0.5		0		0.095	
QNM split (proportion)	0	0	0		0		0.33		0.048	

+ includes equivocal qMSE outcomes equal to the count of icon